

FINAL TECHNICAL REPORT

A Reproducible Rust Ladder from CC(8) to 451 Million Determinants

Quantum Harness Challenge #129 / Team Ranger
Chenxi Wan, Yedi Shen, Junkai Wang
Submission date: 2026-07-30

PRIMARY	LARGEST CONVERGED	HPC EVIDENCE
245,025 determinants	451,681,246 determinants	560 CPUs observed

Outcome

The mandatory Level 0-4 challenge is complete. The checked Rust implementation covers determinant construction, matrix-free FCI, arbitrary-rank CC through CC(8), CI, MBPT, UCC, direct libcint integrals, Rust RHF/DIIS, symmetry sectors, multi-root Davidson, deterministic parallel sigma, and restartable disk checkpoints. All 36 published primary entries match at the source paper's printed precision.

This PDF is a compact submission artifact. The repository's Markdown reports, JSON fixtures, raw logs, checksums, source, tests, and verifier remain authoritative.

1. Acceptance result

The primary Hamiltonian is H2O/6-31G with the oxygen 1s orbital frozen, 12 active spatial orbitals, eight active electrons, and 245,025 determinants. Matrix-free Davidson FCI converges to -76.121174204141965 Eh.

Required family	Delivered	Validation
CC(n)	CC(1)-CC(8)	8/8 published entries match
CI(n)	CI(1)-CI(8)	8/8 published entries match
MBPT(n)	MBPT(1)-MBPT(20)	20/20 published entries match
FCI	Dense and matrix-free Davidson	Small independent oracles + primary residual
Direct integrals	libcint -> Rust RHF -> AO-to-MO -> FCI	Shared FCI engine and fixtures

Acceptance interpretation

Basic requirement completion is 100%, not a forecast. Relative to only the mandatory acceptance ladder, the project also delivers substantial extensions. A numeric 'over-completion percentage' is not an official score, so this report lists extra capabilities instead of inventing grading points.

Representative primary values

Quantity	Value
FCI	-76.121174204141965 Eh
CC(2) / CCSD	-76.119629519205702 Eh
CC(8)	-76.121174196144139 Eh
CC(8) - FCI	7.998e-09 Eh

2. Largest completed exact sector

The largest converged run is all-electron H2O/cc-pVDZ in the exact C2v/A1 ground-state block. Point-group block diagonalization changes representation size, not the Hamiltonian or determinant completeness inside A1.

Quantity	Recorded value
Spatial orbitals / electrons	24 spherical / 10 all electron
Determinants in C2v/A1	451,681,246
Reported FCI energy	-76.24321859 Eh
Residual norm	6.602e-08
Davidson iterations	21
Wall time	03:55:43
Requested memory	384 GiB
Scheduler MaxRSS	222.257 GiB, transcribed; raw accounting unavailable

Exact scope boundary

The symmetry-free representation contains 1,806,590,016 determinants and was bounded and kernel-benchmarked, but it was not solved to convergence. The public eight-decimal energy is scientifically accepted within the recorded residual. The exact production direct_fci.rs and raw Slurm accounting row are not archived, so production provenance is explicitly incomplete.

3. Algorithmic and engineering contributions

These are project contributions and implementation advances. They should not be described as field-wide novel inventions without a separate literature novelty review.

Exact ranked subset-convolution CC exponential

Computes $\exp(T)|\text{HF}\rangle$ by ranked subset convolution and supports arbitrary determinant CC(n) through CC(8). A finite Taylor implementation remains a small-system oracle. This removes the previous all-amplitude exponential bottleneck and makes full-rank validation practical.

Matrix-free spin-free sigma FCI

Applies the Hamiltonian without materializing the determinant-space matrix. Restarted Davidson, independent diagonals, memory preflight, and checked index arithmetic allow the same core to span tiny dense oracles and very large sectors.

Deterministic bounded-memory parallel sigma

Uses fixed source blocks, thread-local partial vectors, and an ordered reduction. It trades some memory and peak speed for reproducibility. The Apple M4 primary benchmark measured a 3.236817x median serial-to-parallel time ratio.

Symmetry propagated through the method stack

Compact Abelian point-group sector enumeration is carried through FCI, CI, MBPT, CC, and UCC. For cc-pVDZ it reduces the exact target representation from 1.806 billion to 451.681 million determinants, a fourfold reduction without discarding an A1 determinant.

Robust Davidson infrastructure

Versioned disk checkpoints use atomic state updates, input/configuration hashes, corruption rejection, and safe resume. Multi-root block Davidson and general active-space combinadic rank/unrank extend the solver beyond one ground-state fixture.

Fail-closed evidence design

Machine-readable evidence records accepted quantities and rejected claim boundaries. The final verifier recomputes hashes and refuses to turn requested resources, unverified accounting, or future methods into completed results.

4. HPC evidence and efficiency

SCNet validates portability, robustness, deterministic repeated solves, and ensemble throughput. It does not demonstrate MPI scaling of one eigenproblem.

Evidence	Result
Robustness grid	18/18 converged
Repeated solves	216/216 converged
Repeated case determinism	true
Observed allocation peak	560 CPUs across 10 tasks
Requested but not observed	1,008 CPUs
Evidence files / manifests	960 / 37

Optimization conclusion

For the 245,025-dimensional case, effective busy cores saturated well below a 56-CPU allocation. Packing independent 14-thread processes is therefore the correct throughput optimization for the submitted ensemble. For one much larger FCI solve, exact point-group reduction and memory-aware Davidson storage provide more value than simply requesting more cores.

5. Reproduction and artifact map

Run the complete local acceptance gate:

```
uv sync --locked
scripts/verify-submission.sh
```

Run the lightweight evidence audit:

```
python3 scripts/hpc/verify_final_evidence.py
```

Artifact	Path
Narrative report	reports/final-competition-summary.md
This PDF	output/pdf/quantum-harness-129-final-technical-report.pdf
Plain-text results	output/data/quantum-harness-129-final-results.txt
Large FCI JSON	fixtures/h2o-ccpvdz-ae/fci-c2v-xh5-result.json
SCNet JSON	fixtures/hpc/scnet-2026-07-30.json
Reproduction prompt	docs/reproduction-prompt.md
Evidence verifier	scripts/hpc/verify_final_evidence.py

Public endpoints

Repository: <https://github.com/JunkaiWang-TheoPhy/quantum-harness-129-workbench-rust>

Upstream PR: <https://github.com/QuantumBFS/quantum.harness/pull/217>

Evidence snapshot: 720307ad6ecef3a39b2135ccf7ad53c2962f12ab

5-hour-to-finish lesson

Freeze numerical evidence first; run only missing bounded checks locally; use HPC only for already-prepared, auditable workloads; write claims directly from machine-readable artifacts; then generate the PDF, manifest, and PR index from the same frozen evidence. Do not start a new billion-dimensional calculation during submission packaging.

6. Claim ledger and next research step

Claim	Status
Mandatory Level 0-4 challenge	Complete
36 primary published entries	36/36 match at printed precision
C2v/A1 451M determinant FCI	Converged and accepted
Symmetry-free 1.806B determinant FCI	Not converged; bounded benchmark only
Observed thousand-CPU run	Not observed; 560-CPU peak is public
Single-solve MPI scaling	Not implemented
HCI/iCI, EN-PT2, orbital optimization	Future work; not implemented
Quantum advantage	Not claimed

Recommended next algorithm

Add a common selected-determinant interface, deterministic heat-bath or iCI-style selection, a variational solve, EN-PT2 with an explicit error budget, and threshold extrapolation. Then compare natural and orbital-optimized bases on stretched water. The current exact solver supplies the calibration oracle and makes those future approximations scientifically testable.

Submission decision

Submit the existing corrective branch and PR. The basic challenge is already complete; the remaining work is packaging, CI confirmation, and reviewer communication. Avoid adding unverified claims or moving the immutable v0.5.0 tag.